EX27 · CLASS_17_VECTORS · 04 / PSEUDOCODE GUIDE

Think before coding

Transform an algebraic or physical statement into algorithm steps.

Learning sequence
01 Math theory → 02 Python reference → 03 Exercise → 04 Pseudocode → Programming.

Prerequisites: EX25 and EX26, NumPy arrays, Euclidean norm and dot product.

1. Why write pseudocode?

Pseudocode is not Python: it describes **what should happen** in order, before syntax details. It separates inputs, mathematics, validation and outputs.

2. Five-question method

Question	Application to EX27
What is given?	Two vectors and, for physics, units.
What do I need?	Cross product, magnitude, direction.
What equation?	u × v or r × F, depending on the task.
How do I validate?	Manual example, perpendicularity, signs.
What do I print?	Vector, magnitude and units when relevant.

3. Guided pseudocode example (basis vectors only)

This example covers only the Cartesian unit-vector demonstration, not the main tasks.

```
START
  Define vector a with components (1,0,0)
  Define vector b with components (0,1,0)
  Compute c = cross_product(a,b)
  Display c and compare against (0,0,1)
END
```

4. Your design: algebra

Complete this template before coding. Each line must reflect a logical operation. Remember that reversing the order flips the sign.

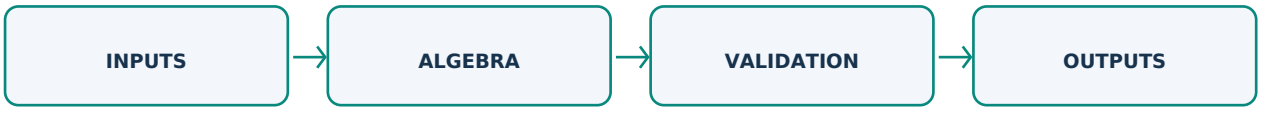
```
START
  1. DEFINE: u = (...), v = (...)
  2. COMPUTE: _____________
  3. COMPUTE: _____________ (reverse order)
  4. COMPUTE: _____________ (norm)
  5. CHECK: _____________ (dot product)
  6. DISPLAY: _____________
END
```

5. Your design: engineering problem

Translate the moment definition into operations. First identify origin O, application point P and units.

```
START
  1. Define the reference point for the moment
  2. Define r from reference point to force
  3. Define F with its three components
  4. Compute M = _____________
  5. Compute magnitude = _____________
  6. Interpret sign and units
  7. Display results
END
```

6. General logic flow



This flow has no branching or loops. Once we introduce decisions (if/else) and repeated calculations, we will add diamonds, arrows and labeled conditions.

7. Pre-run checklist

- Do arrays have three components ordered x,y,z?
- Did I preserve cross-product argument order?
- Did I calculate and validate the resulting magnitude?
- Do the physical sign, reference and units make sense?
- Can each pseudocode step become a line or small Python block?